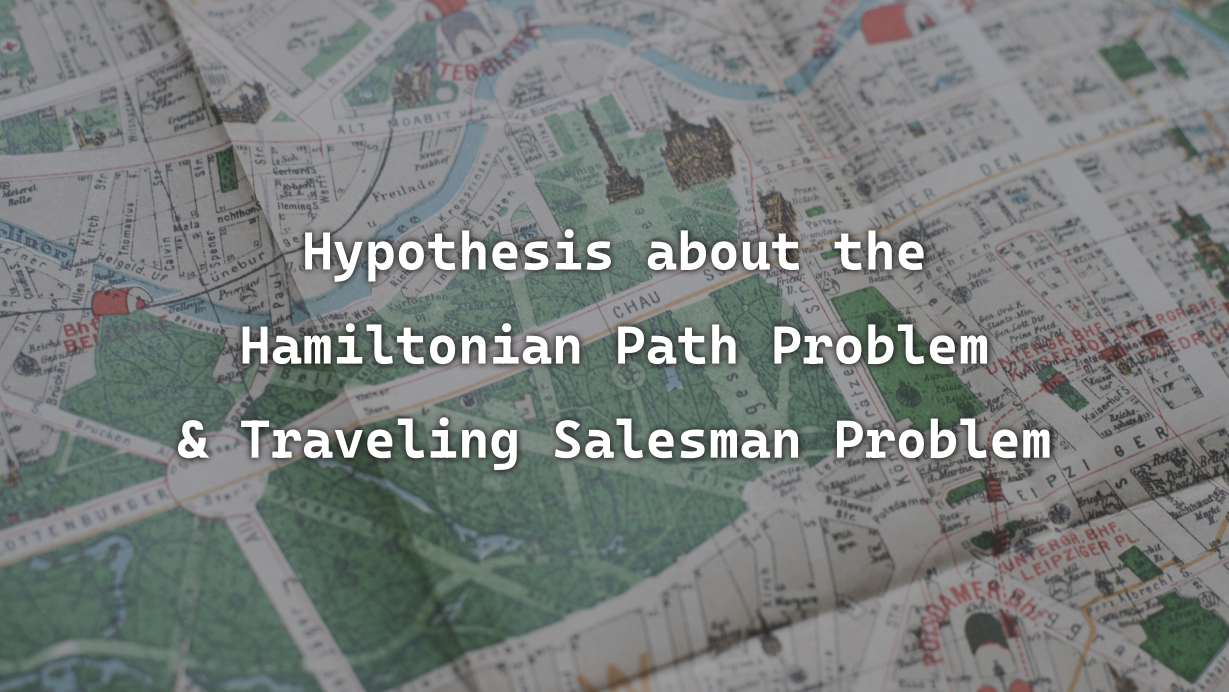
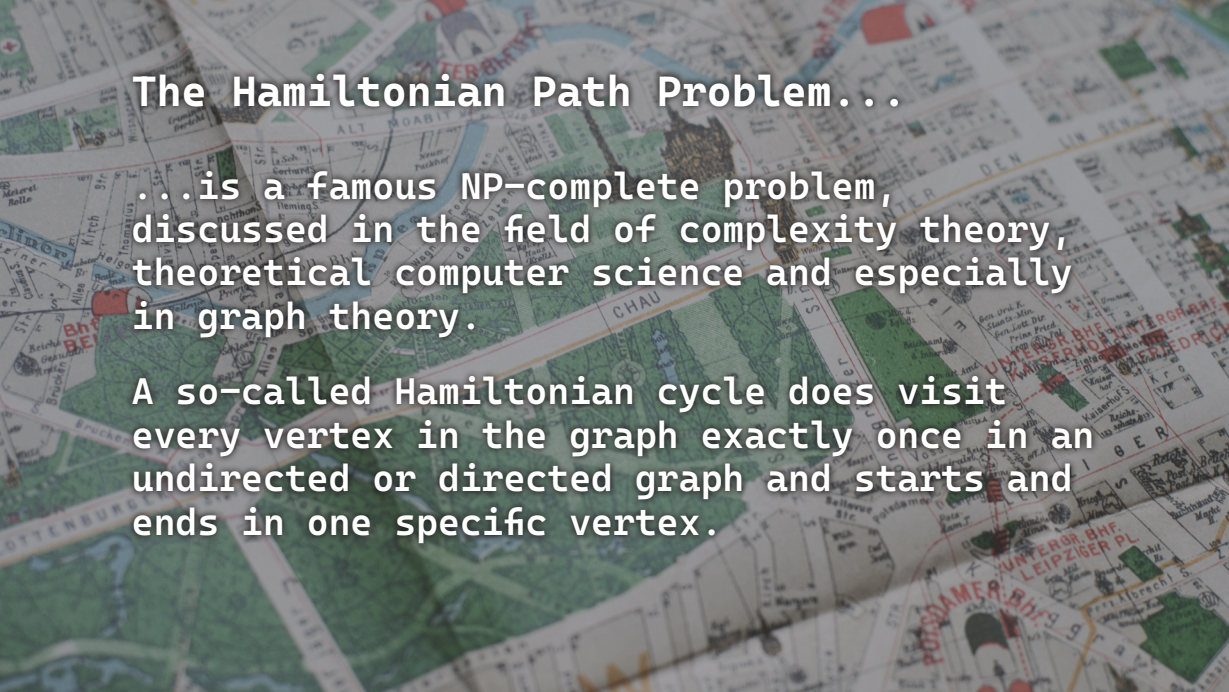




Hypothesis about the Hamiltonian Path Problem & Traveling Salesman Problem



The Hamiltonian Path Problem...

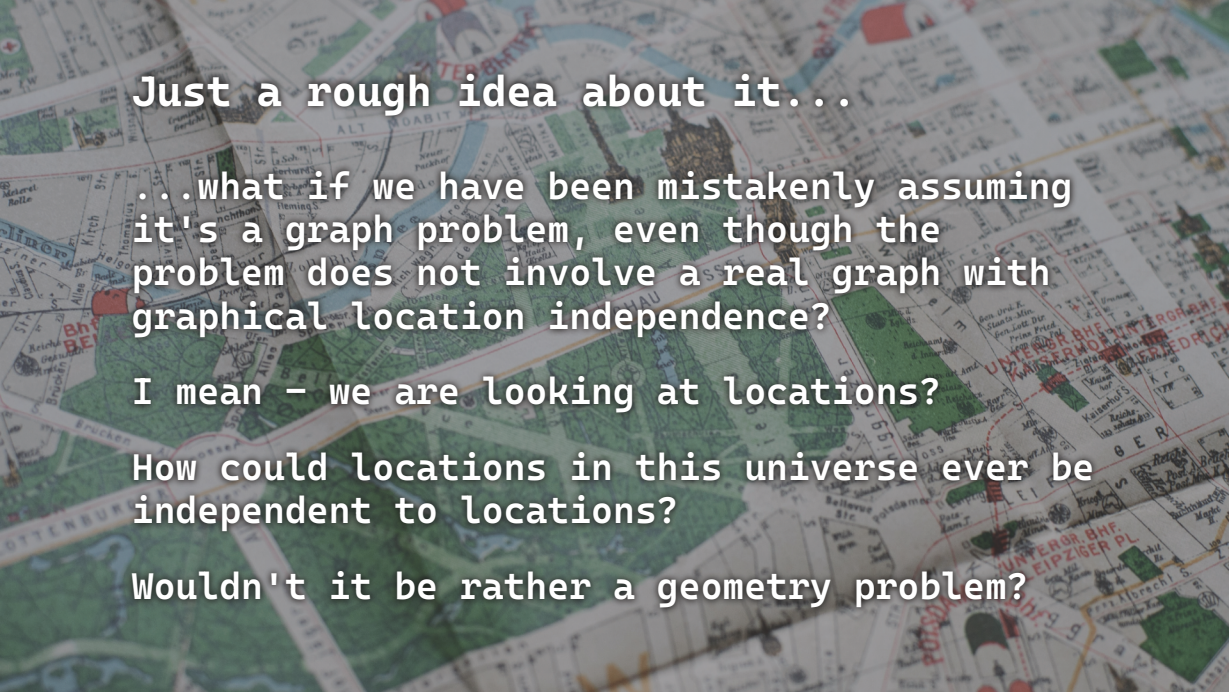
...is a famous NP-complete problem, discussed in the field of complexity theory, theoretical computer science and especially in graph theory.

A so-called Hamiltonian cycle does visit every vertex in the graph exactly once in an undirected or directed graph and starts and ends in one specific vertex.

The Traveling Salesman Problem...

...is a subform of the Hamiltonian path or better Hamiltonian cycle problem and describes a salesperson, who needs to find the shortest path between a number (n) of locations.

A graph is apparently a useful model of a network of locations in so far theory and therefore used in mathematics to look at possible paths.



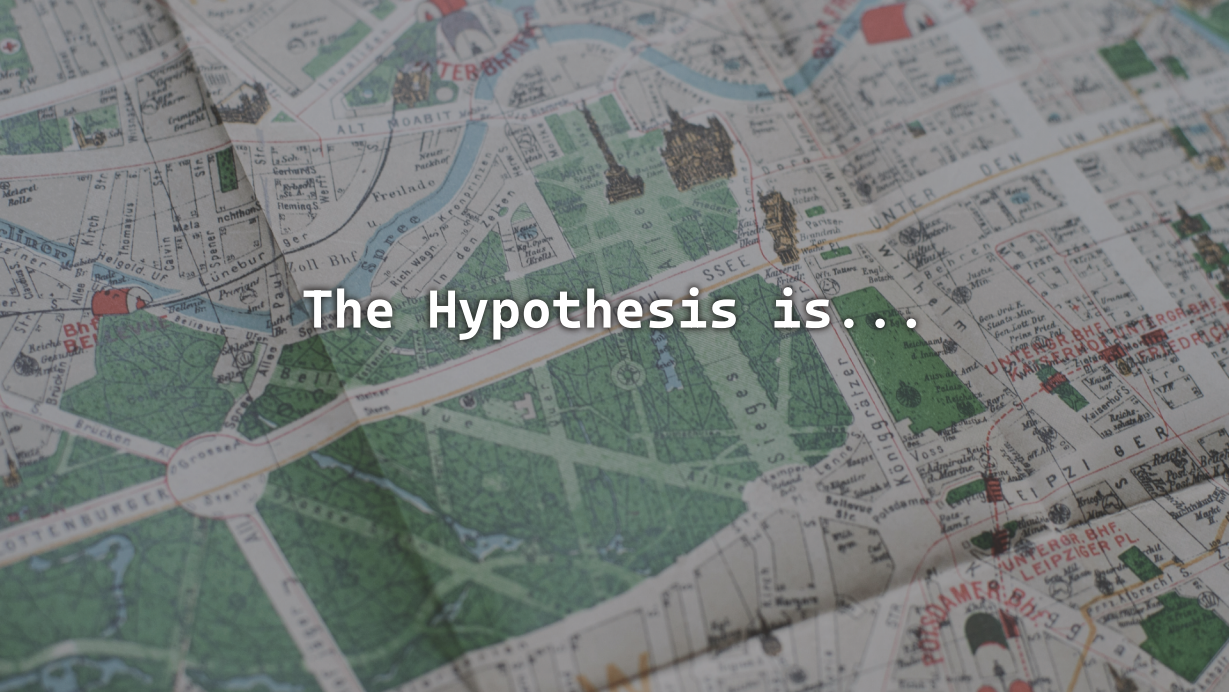
Just a rough idea about it...

...what if we have been mistakenly assuming it's a graph problem, even though the problem does not involve a real graph with graphical location independence?

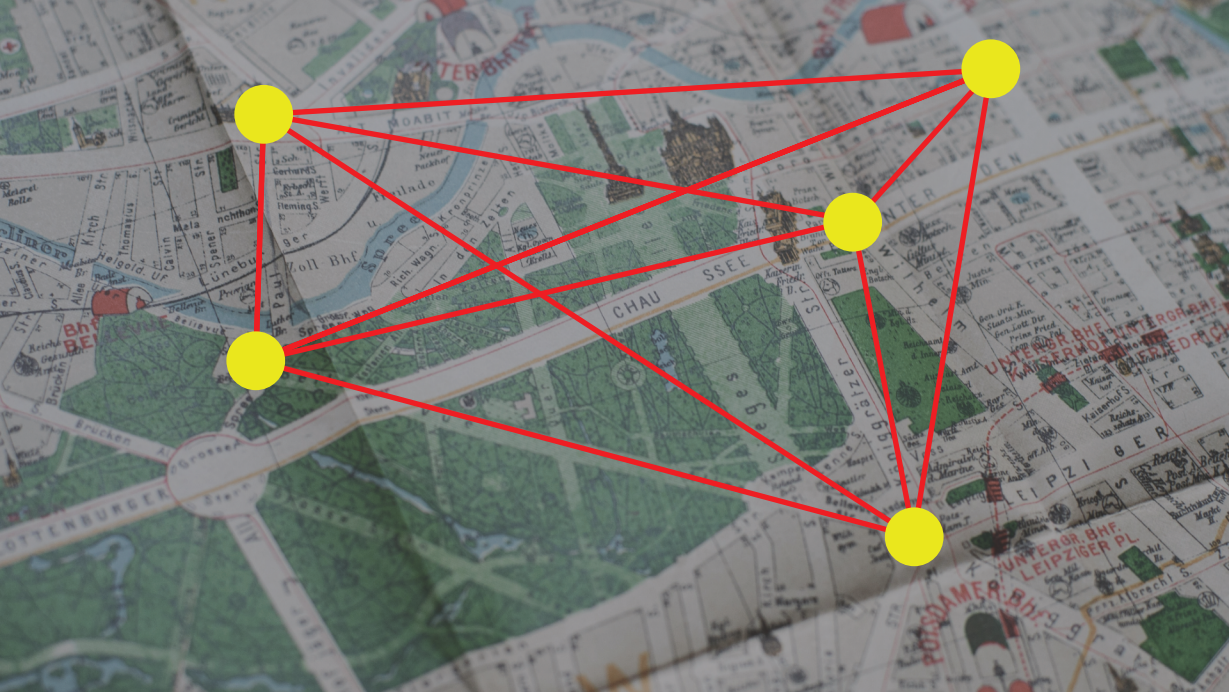
I mean - we are looking at locations?

How could locations in this universe ever be independent to locations?

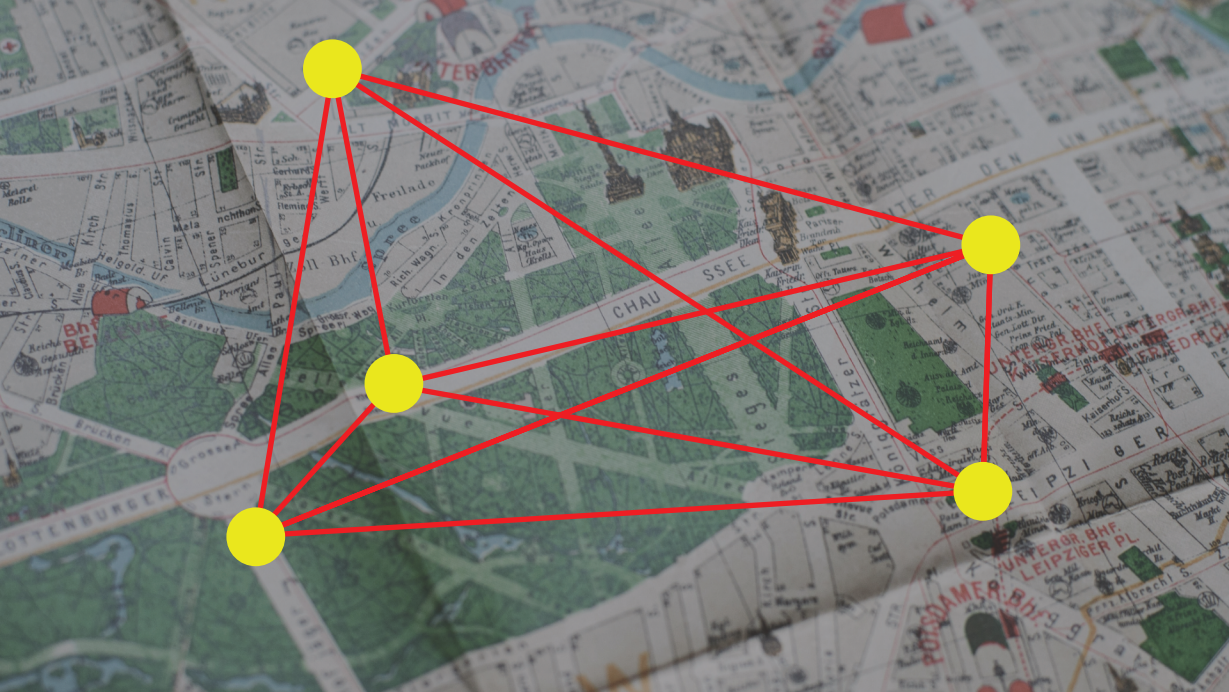
Wouldn't it be rather a geometry problem?



The Hypothesis is...



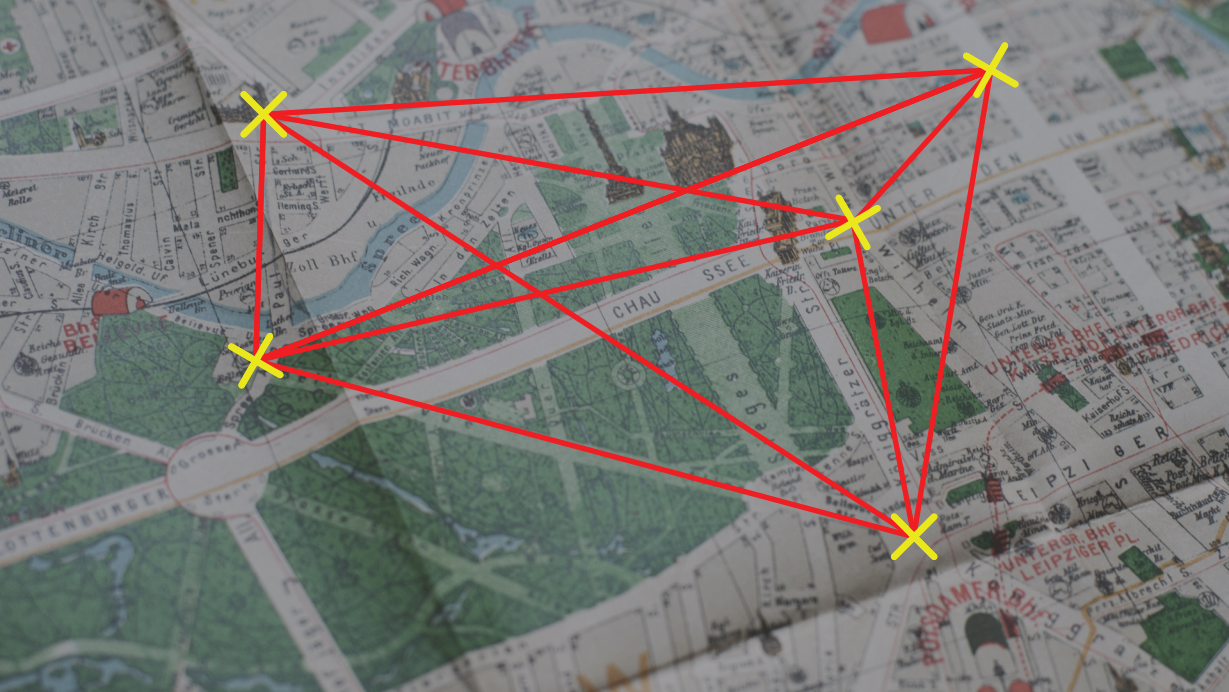


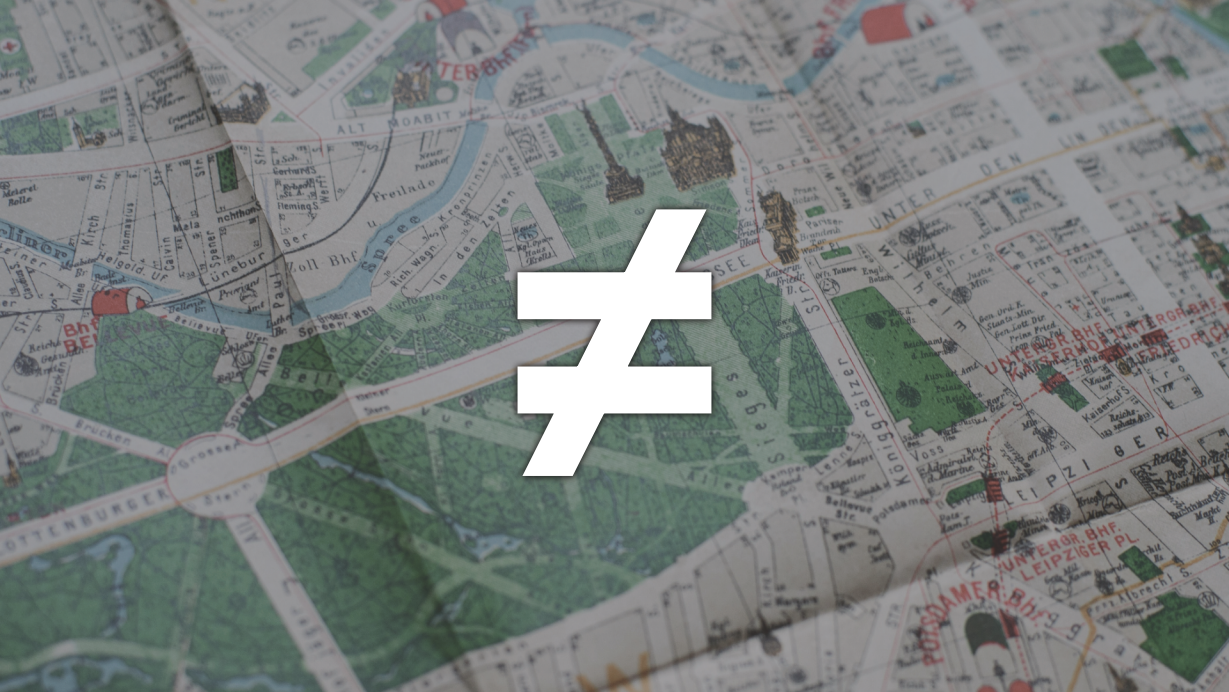


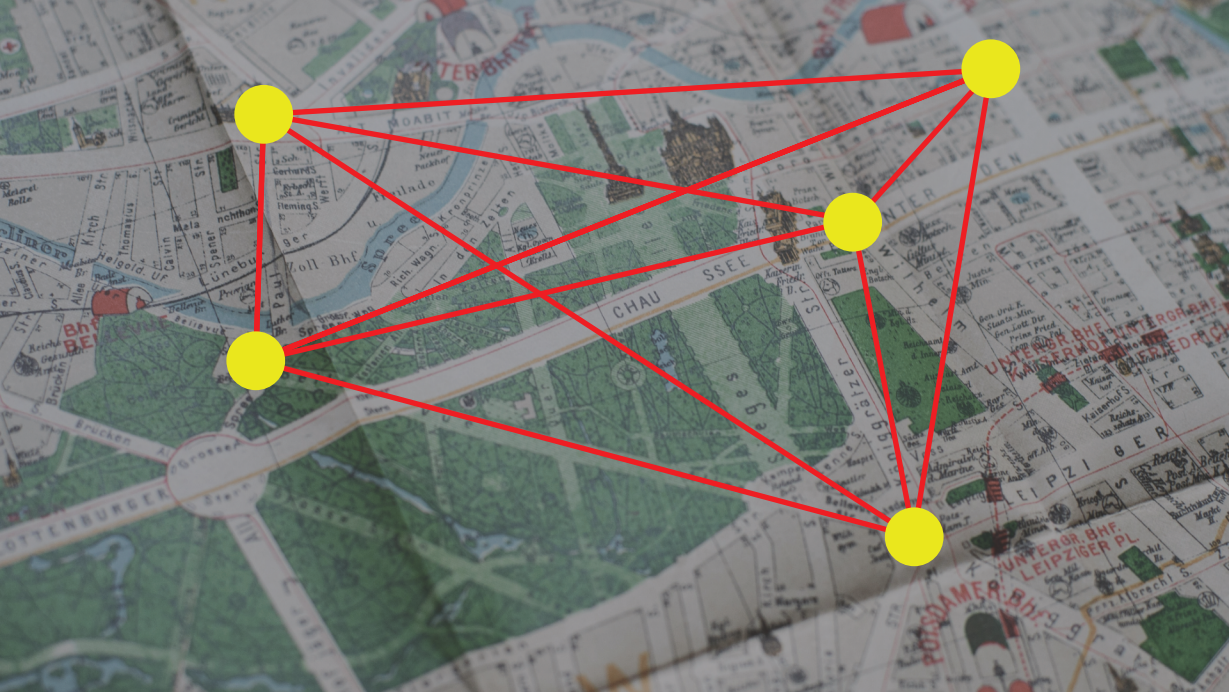


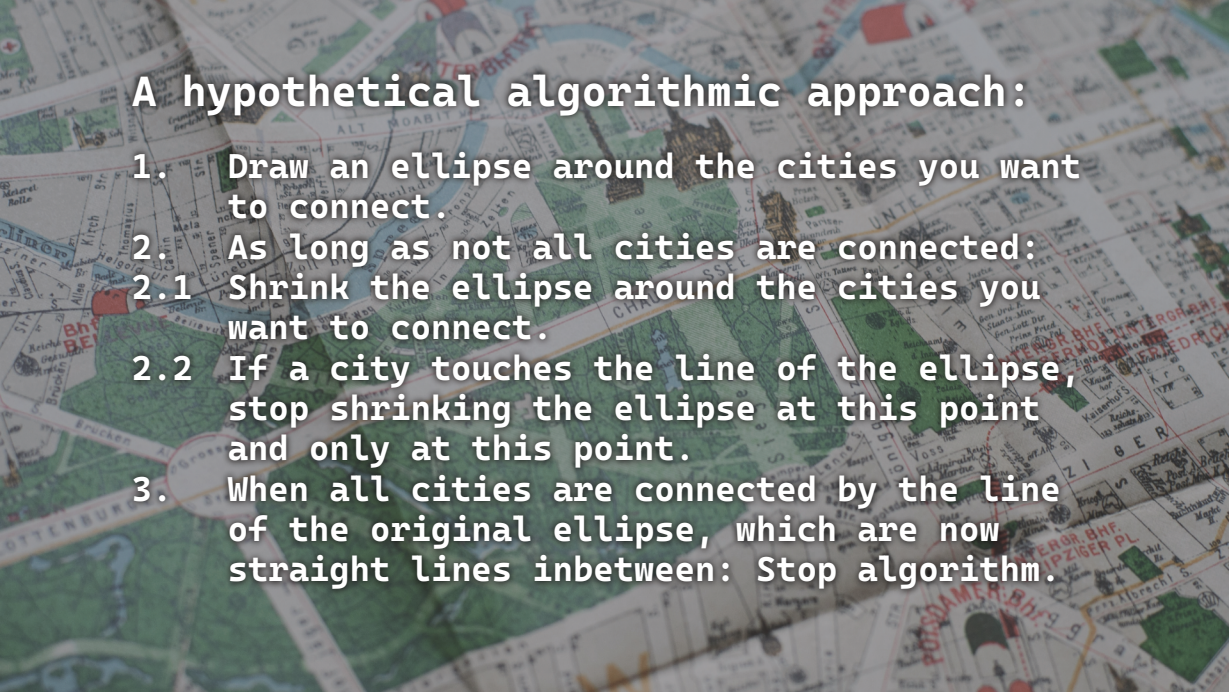












A hypothetical algorithmic approach:

1. Draw an ellipse around the cities you want to connect.
2. As long as not all cities are connected:
 - 2.1 Shrink the ellipse around the cities you want to connect.
 - 2.2 If a city touches the line of the ellipse, stop shrinking the ellipse at this point and only at this point.
3. When all cities are connected by the line of the original ellipse, which are now straight lines inbetween: Stop algorithm.



The Algorithm is...





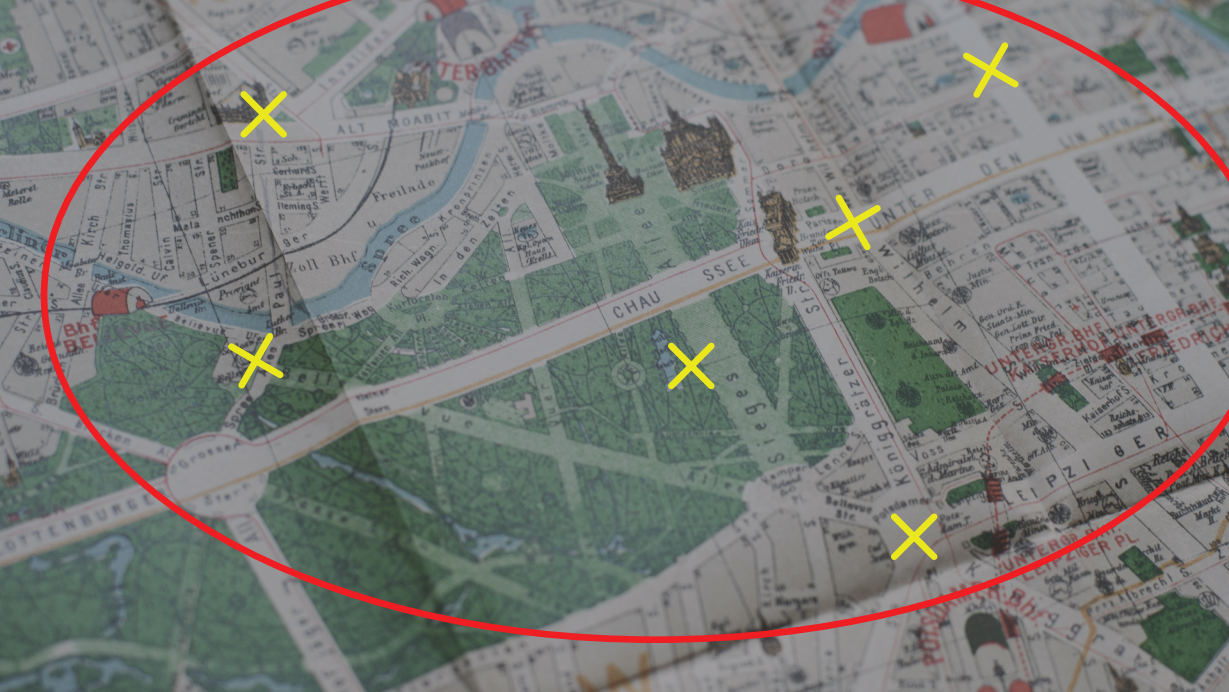






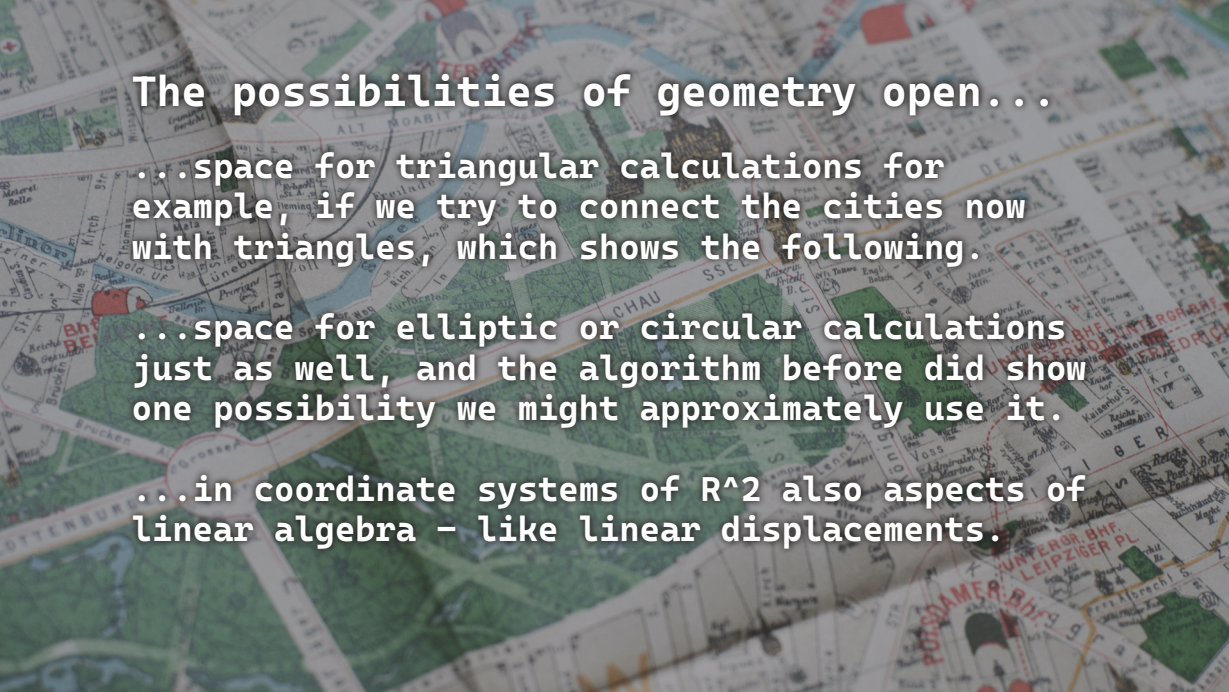










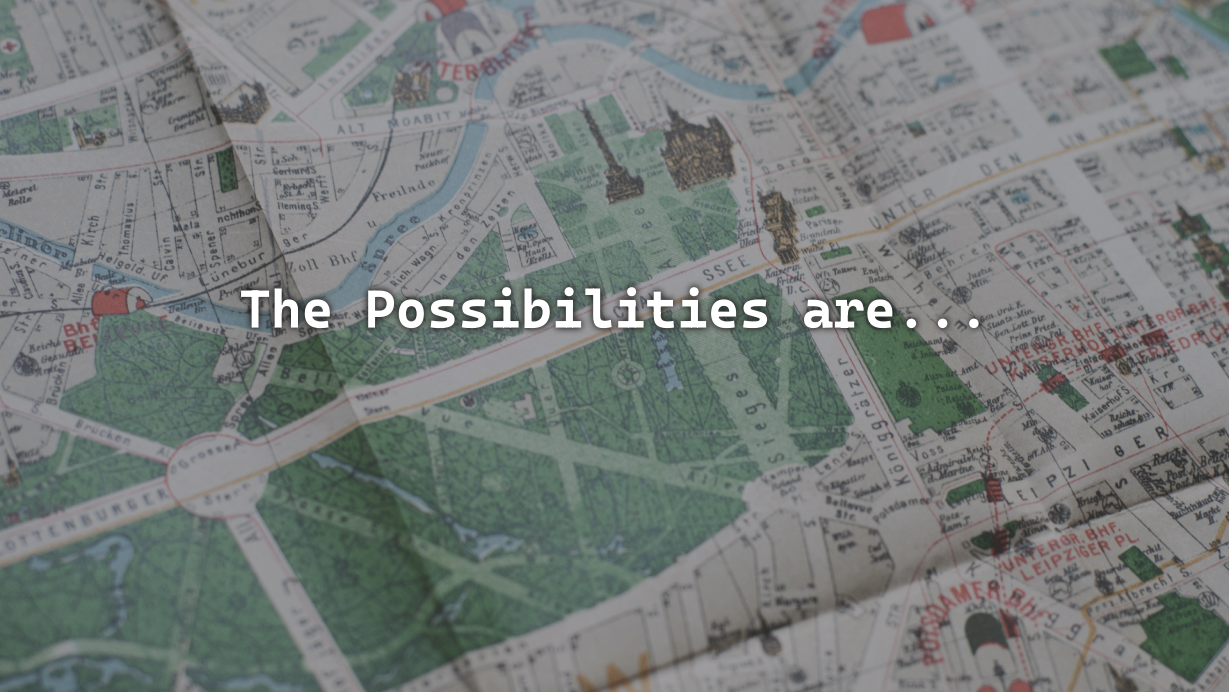


The possibilities of geometry open...

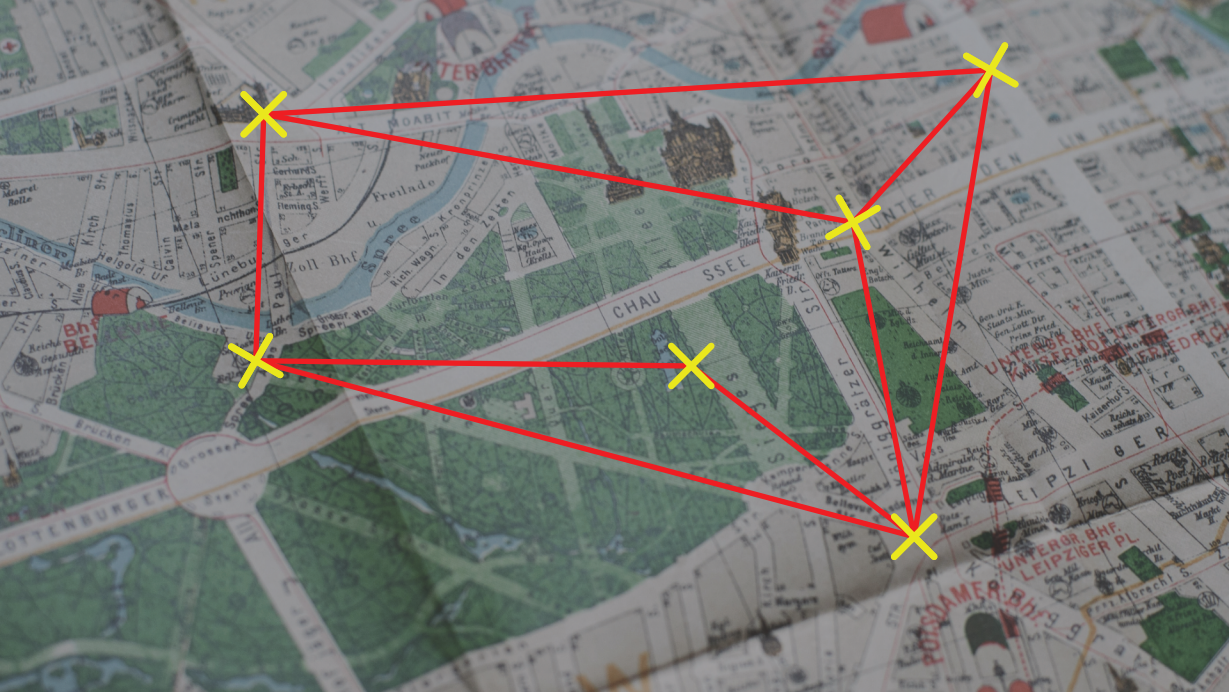
...space for triangular calculations for example, if we try to connect the cities now with triangles, which shows the following.

...space for elliptic or circular calculations just as well, and the algorithm before did show one possibility we might approximately use it.

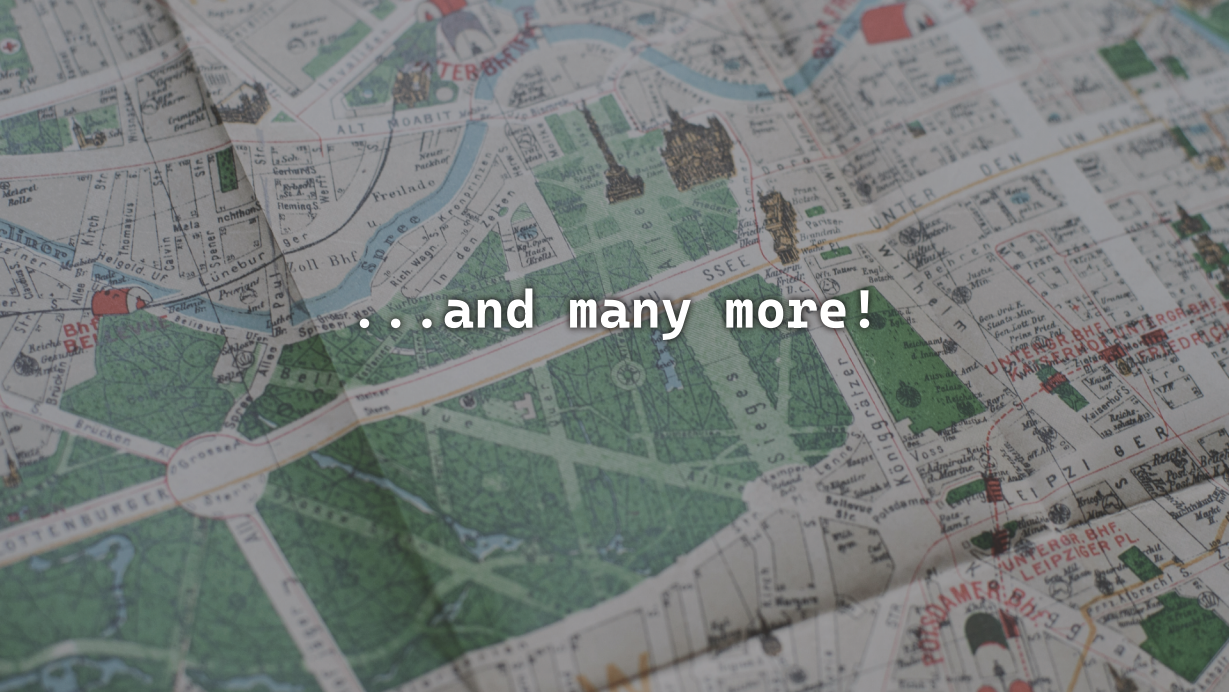
...in coordinate systems of R^2 also aspects of linear algebra – like linear displacements.



The Possibilities are...







...and many more!



Idea & Presentation by Marc-Michel Münch

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